EIGENVALUES & EIGENVECTORS — ENGINEERING MATHEMATICS CHEATSHEET

Definitions • Characteristic equation • Computation methods • Diagonalization • Applications

1. Fundamental Definitions and Formulae

Concept	Formula / Definition	Key note
Eigenvalue–eigenvector	$A\,v = \lambda\,v$, where $v \neq 0$	A is n×n; λ scalar
Characteristic equation	$\det(A - \lambda I) = 0$	Roots are eigenvalues
Characteristic polynomial	$p(\lambda) = \det(\lambda I - A)$	Equivalent roots
Eigenvector for λ	Solve $(A - \lambda I)v = 0$	Nonzero nullspace vector
Eigenspace	$E\lambda = \text{Null}(A - \lambda I)$	Includes zero vector; eigenvectors are nonzero members
Algebraic multiplicity	Multiplicity of λ as root of $p(\lambda)$	AM
Geometric multiplicity	$\dim \text{Null}(A - \lambda I)$	GM; $1 \le GM \le AM$
Trace / determinant	$\Sigma\,\lambda_i = \text{tr}(A)$; $\Pi\,\lambda_i = \det(A)$	Count multiplicities
Similarity transformation	$B = P^{-1}AP$	Same characteristic polynomial
Diagonalization	$A = PDP^{-1}$ (equiv. $P^{-1}AP = D$)	P columns are eigenvectors

2. Useful Results and Special Cases

Matrix / condition	Result	Interpretation
Triangular matrix	Eigenvalues = diagonal entries	Upper or lower triangular
Real symmetric A	All eigenvalues real; orthogonal eigenvectors for distinct eigenvalues	$A^T = A$
Orthogonal diagonalization	$A = QDQ^T$; $Q^TQ = I$	For real symmetric matrices
Singular matrix	0 is an eigenvalue iff $\det(A)=0$	Non-invertible
A invertible	Eigenvalues of A^{-1} are $1/\lambda_i$	$\lambda_i \neq 0$
A^k	Eigenvalues λ_i^k (same eigenvectors)	For integer $k \ge 0$
cA	Eigenvalues $c\lambda_i$	Eigenvectors unchanged if $c \neq 0$
A + cI	Eigenvalues $\lambda_i + c$	Eigenvectors unchanged
Distinct eigenvalues	Corresponding eigenvectors are linearly independent	Sufficient for diagonalization

3. Cayley–Hamilton Theorem

- Every square matrix satisfies its own characteristic equation: if $p(\lambda)=\det(\lambda I-A)=\lambda^n+c_1\lambda^{n-1}+\ldots+c_n$, then $p(A)=A^n+c_1A^{n-1}+\ldots+c_nI=0$.
- Use: reduce high powers of A to lower powers; obtain A^{-1} for nonsingular A by rearranging $p(A)=0$.
- For 2×2 A: $A^2 - (\text{tr } A)A + (\det A)I = 0$.

EIGENVALUES & EIGENVECTORS — METHODS AND PROCEDURES

A reliable workflow for eigenvalue computation, eigenvectors, and matrix diagonalization

4. Method A — Find Eigenvalues and Eigenvectors

Step	Solution procedure
1. Form matrix	Write A and identify its order n.
2. Characteristic matrix	Construct $A - \lambda I$ (subtract λ from every diagonal entry).
3. Determinant	Compute $\det(A - \lambda I)$ and set it equal to zero.
4. Solve polynomial	Find all roots $\lambda_1,\ldots,\lambda_n$, including multiplicities.
5. Eigenvector system	For each λ_i, solve $(A - \lambda_i I)v = 0$ by row reduction.
6. State eigenspace	Give a nonzero vector (or basis) and verify $A\,v = \lambda v$.

5. Method B — Diagonalization

Step	Procedure / test
1. Find spectrum	Calculate eigenvalues and eigenvectors.
2. Build P	Place n linearly independent eigenvectors as columns of P.
3. Build D	Place corresponding eigenvalues on diagonal in the same column order.
4. Check independence	$\det(P) \neq 0$, or verify rank(P)=n.
5. Conclude	$A = PDP^{-1}$, or $P^{-1}AP = D$.
Diagonalizable criterion	A is diagonalizable iff it has n linearly independent eigenvectors; equivalently, for each eigenvalue, GM = AM and total eigenvector dimension is n.

6. Worked Example — 2 × 2 Matrix

Step	Calculation
Given	$A = [[2, 1], [1, 2]]$
Characteristic equation	$\det(A - \lambda I) = (2-\lambda)^2 - 1 = 0$
Factor	$(2-\lambda-1)(2-\lambda+1)=0 \Rightarrow \lambda_1=3, \lambda_2=1$
For $\lambda=3$	$A-3I = [[-1,1],[1,-1]] \Rightarrow x=y$; choose $v_1=[1,1]^T$
For $\lambda=1$	$A-I = [[1,1],[1,1]] \Rightarrow x=-y$; choose $v_2=[1,-1]^T$
Diagonal form	$P=[[1,1],[1,-1]]$, $D=[[3,0],[0,1]]$; $A=PDP^{-1}$
Verification	$A[1,1]^T=3[1,1]^T$; $A[1,-1]^T=[1,-1]^T$

7. Quick Checks and Common Pitfalls

- Use $\det(A-\lambda I)=0$; do not confuse determinant with solving $(A-\lambda I)v=0$.
- Eigenvector must be nonzero; any nonzero scalar multiple is also an eigenvector.
- Keep eigenvalue order consistent between columns of P and diagonal entries of D.
- Repeated eigenvalues do not automatically imply diagonalizability; compare AM and GM.
- Check candidate vectors by direct multiplication $A\,v = \lambda v$.
- For symmetric real matrices, use orthonormal eigenvectors to form Q and write $A=QDQ^T$.
- Do not infer eigenvalues from diagonal entries unless A is triangular (or another applicable structure is established).